STAFF REVIEW · H446 199 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. C.

2. Last In, First Out.

3. The array must be sorted.

4. B, before changing A.next to X.

I Do — source worked steps

1. Predict — Compare 12 with 20 → go left to 10.

2. Confirm and explain — 12 > 10, so 12 becomes the right child of 10.

We Do — guided CHECK

1. Confirm and explain
35 is smaller than 40, so insert it as the left child of 40.

You Do — source mark schemes

1. Left child of 75 because 60>50 and 60<75.

2. 18 is the left child of 20, which is the right child of 15.

3. A sequence of sorted inserts can make it a chain of n nodes; each lookup may visit every node.

Exit Ticket — source answer

1. False. An unbalanced BST may degenerate into a chain with O(n) lookup.